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Modern Cipher and Module Objectives

Modern Cipher and Block Cipher (vs. Stream Cipher)

Ideal Block Cipher

Feistel Cipher

Data Encryption Standard (DES)

Video: DES Overview
3 min

Video: DES Round Function
1 min

Reading: About Pseudocode
30 min

Peer-graded Assignment: DES Overview Pseudo-Code Grading in progress

Review Your Peers: DES Overview Pseudo-Code

Video: DES Subkey Generation
1 min

Video: DES Security
4 min

Reading: Lecture Slides for Block Cipher and DES
15 min

Quiz: DES
Submitted

Peer-graded Assignment: DES Overview Pseudo-Code

Reviews 3 left to complete

DES algorithm

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July 7, 2024

Translate this submission

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PROMPT	RUBRIC
Which programming language, if any, are you familiar with? Java, C, Perl, Python	Are you (the reviewer) familiar with the programming language that the submitter identified in his/her answer? (The submitter receives credit regardless.) 1 pt No 1 pt Yes

PROMPT	RUBRIC
Write your answer for the pseudo-code for the DES encryption. <pre>//INPUT - input 64 bit to encrypt //SUBKEY[] - input 16 elements array with subkeys for all rounds (a total of 768 bits), each subkey is 48 bits</pre> DES start Declare a 64 bit variable OUTPUT Declare a 32 bit variable L Declare a 32 bit variable R Declare a loopcounter OUTPUT = IP(INPUT) //initial permutation L = first 32 bits from OUTPUT //initial split into L and R R = last 32 bits from OUTPUT for loopcounter = 0 to 15 // 16 rounds Declare a 32 bit variable TEMP TEMP = R R = FP(R, SUBKEY[loopcounter]) R = R XOR L L = TEMP end loop Set first 32 bits of OUTPUT to L // combine results, note last swap from the loop took place Set last 32 bits of OUTPUT to R OUTPUT = FP(OUTPUT) // final permutation DES end	Does the pseudo-code correctly describe DES Encryption supporting correct data/key types/lengths? 0 pts Most, if not all, of the pseudo-code components are flawed. 2 pts The answer has some components but requires major overhaul or built something else (not a DES cipher). 4 pts The answer misses a few steps/variables, impacting its logic, or only works in special circumstances (e.g., restrictions on variables). 6 pts The answer misses a step/variable but it is minor, and the pseudo-code still implements DES Encryption or the error can be corrected quickly. 8 pts The answer provides a correct DES encryption supporting the correct input/output and arbitrary-length of key. How is the pseudo-code presentation? 0 pts I cannot follow the logic of the pseudo-code. 1 pt The pseudo-code is so long that it is distracting; and the documentation is poor. 2 pts The pseudo-code is hard to follow but is understandable. Or the pseudo-code is unnecessarily long. 3 pts The pseudo-code is generally easy to follow but lacks some documentation. 4 pts The pseudo-code is easy to follow with clear definitions and explanations/documentation.

OVERALL ASSIGNMENT RUBRIC
Does the pseudo-code correctly describe DES cipher operation? 0 pts Most, if not all, of the pseudo-code components are flawed. 2 pts The answer has some components but requires major overhaul, or the answer built another cipher. 4 pts The answer misses a few steps/variables, impacting its logic, or only works in special circumstances (e.g., restrictions on variables). 6 pts The answer misses a step/variable but the error is relatively minor and can be corrected quickly. 8 pts The answer provides a correct DES encryption supporting the correct data and key. How is the pseudo-code presentation? 0 pts I cannot follow the pseudo-code logic. 1 pt The pseudo-code is very difficult to follow, for example, it is so long that it is distracting, or the documentation is poor. 2 pts The pseudo-code is hard to follow but is understandable. Or the pseudo-code is unnecessarily long. 3 pts

